How the Application Provides Confidentiality, Integrity and Sender Authentication

Confidentiality

The exam paper is encrypted with AES-256-GCM using a random session key K. The key itself is never sent as plain data. It is wrapped with the recipient's RSA-3072 public key using OAEP padding with SHA-256, so only the matching private key can recover it. Anyone who captures the data on the channel sees only ciphertext and the wrapped key, never the exam paper or the key.

Integrity

Two checks catch any change to the data.

1. AES-GCM produces a 128-bit authentication tag over the ciphertext. If a single bit of the ciphertext changes, decryption fails.
2. Every package sent, the session key package and each file package, is signed with RSA-PSS and SHA-256 over the full package, not just the ciphertext. This covers the session id, nonce, ciphertext, tag, and metadata together, so a change to any one of these fields breaks the signature check even if it would not have broken the GCM tag on its own.

Sender authentication

The RSA-PSS signature is created with the sender's private key and checked against the sender's public key, which both sides already hold through a trusted exchange. A signature that passes this check could only have been produced by someone holding that private key, so it proves the package came from the claimed sender and not an impersonator.

In all three areas the application fails closed. If either the signature check or the authentication tag check does not pass, the package is rejected and no plaintext is released.